

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

28/05/2026, 12:05

1. **Value of a_2 :** The code defines $a_2 = \lim_{n \rightarrow \infty} (2f(n) + \frac{1}{2} + 0.25^n)$.

Inside $f(n)$, the initial term and the subtracted geometric series involve 72^{-N} with $N = 10^{24}$, which is astronomically tiny.

The limit evaluates to $a_2 = 1 + \varepsilon$, where $\varepsilon = 2 \left(72^{-N} - \frac{72^{-3N}}{1-72^{-N}} \right) > 0$ is an extremely small positive number (effectively $a_2 \approx 1.000 \dots$).

2. **Optimisation problem:** Minimise $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$.

Since $a_2 > 1$, total time $T = 1 + (1 - a_2)x_2$ decreases as x_2 increases. The unique optimal point is $(x_1^*, x_2^*) = (0, 1/a_2) \approx (0, 0.999 \dots)$.

3. **Infinity-norm distances:**

- For $(0.3, 0.7)$: distance = $\max(|0.3 - 0|, |0.7 - 1/a_2|) = \max(0.3, 0.3 - \frac{\varepsilon}{1+\varepsilon}) = 0.3 \leq 0.4$. True.
- For $(0.6, 0.4)$: distance = $\max(0.6, |0.4 - 1/a_2|) = \max(0.6, 0.6 - \frac{\varepsilon}{1+\varepsilon}) = 0.6 \not\leq 0.45$. False.